

School of Engineering and Applied Science

Lecture Scribe: Lecture 26 (CSE 400)

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1) List of Topics Covered

1. Basics of Stochastic Processes

- Waveform Collections and Realizations
- Process Characterization (First, Second, and Nth Order Density Functions)

2. Statistical Moments

- Mean Function (Ensemble Average)
- Autocorrelation and Autocovariance Functions
- Correlation Coefficient
- Properties of Autocorrelation

3. Stationary Stochastic Process

- Strict Sense Stationary (SSS) Process
- Wide Sense Stationary (WSS) Process

4. Ergodic Process

- Time Average of a Random Process
- Mean Ergodicity and Autocorrelation Ergodicity

2) Explanation of Topics Covered

1. Basics of Stochastic Processes

A Stochastic Process $X(\zeta, t)$ is a rule for assigning to every outcome ζ of a random experiment a function of time $X(\zeta, t)$. At a fixed time t_k , $X(\zeta, t_k)$ is a random variable. A single time function $x_i(t)$ corresponding to a specific outcome is called a realization or sample function. Characterization involves Nth-order joint probability density functions.

2. Statistical Moments

Since Nth-order PDFs are hard to find, we use moments. The mean function $\mu_X(t)$ is the expected value across the ensemble at time t . The autocorrelation $R_X(t_1, t_2)$ describes the dependence between values at two time instants. Autocovariance $C_X(t_1, t_2)$ removes the mean effect to look at the correlation of the fluctuations.

3. Stationary Processes

A process is stationary if its statistical properties are invariant to time shifts. SSS requires the joint PDF to remain the same regardless of time origin. WSS is a relaxed condition requiring only a constant mean and an autocorrelation that depends solely on the time difference $\tau = t_1 - t_2$.

4. Ergodic Processes

Ergodicity implies that "time averages" equal "ensemble averages." In many applications, we only have one realization of a signal. If the process is ergodic, we can calculate its statistics (mean, variance, etc.) by averaging a single sample over a long enough time interval.

3) List of Definitions and Theorems

1. Definition: Mean Function

The mean of a random process $X(t)$ is defined as the ensemble average:

$$\mu_X(t) = E[X(t)] = \int_{-\infty}^{\infty} x f_{X(t)}(x) dx$$

2. Definition: Autocorrelation Function

The autocorrelation $R_X(t_1, t_2)$ of a random process $X(t)$ is:

$$R_X(t_1, t_2) = E[X(t_1)X(t_2)]$$

3. Definition: Wide Sense Stationarity (WSS)

A process $X(t)$ is WSS if:

1. $E[X(t)] = \mu_X$ (Constant)
2. $E[X(t)X(t + \tau)] = R_X(\tau)$ (Depends only on time difference τ)

4. Theorem: Symmetry of Autocorrelation

For a WSS process, the autocorrelation function is an even function: $R_X(\tau) = R_X(-\tau)$.

5. Detailed Step-by-Step Proof

1. By definition of WSS: $R_X(\tau) = E[X(t)X(t + \tau)]$.
2. Let $t + \tau = t'$, then $t = t' - \tau$.
3. Substitute these into the expectation: $R_X(\tau) = E[X(t' - \tau)X(t')]$.
4. By commutativity of multiplication: $E[X(t')X(t' - \tau)]$.
5. Since the process is WSS, this expectation only depends on the difference between the time indices: $t' - (t' - \tau) = \tau$. However, looking at the indices t' and $t' - \tau$, the shift is $-\tau$.
6. Thus, $R_X(\tau) = R_X(-\tau)$.

4) Important Examples

Example 1: Sinusoid with Random Phase

Problem : Consider the process $X(t) = A \cos(\omega_c t + \Theta)$, where A and ω_c are constants and $\Theta \sim U[0, 2\pi]$. Determine if $X(t)$ is WSS.

Solution:

- **Step 1: Check the Mean.**

$$\mu_X(t) = E[A \cos(\omega_c t + \Theta)] = \int_0^{2\pi} A \cos(\omega_c t + \theta) \frac{1}{2\pi} d\theta.$$

The integral of a cosine over its full period is 0. So, $\mu_X(t) = 0$ (Constant).

- **Step 2: Check Autocorrelation.**

$$R_X(t, t + \tau) = E[A \cos(\omega_c t + \Theta) A \cos(\omega_c(t + \tau) + \Theta)].$$

- **Step 3: Use Trigonometric Identity.**

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)].$$

$$R_X(t, t + \tau) = \frac{A^2}{2} E[\cos(\omega_c \tau) + \cos(2\omega_c t + \omega_c \tau + 2\Theta)].$$

- **Step 4: Evaluate Expectation.**

The second term's expectation is 0 because it's a sinusoid over 4π .

$$R_X(\tau) = \frac{A^2}{2} \cos(\omega_c \tau).$$

- **Result:** Since the mean is constant and R_X depends only on τ , the process is **WSS**.

Example 2: Time Averaging for Mean Ergodicity

Problem : For a specific realization $x(t) = A \cos(\omega_c t + \theta)$, find the time average $\bar{x}$.

Solution:

- **Step 1:** Apply the time average formula.

$$\bar{x} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T A \cos(\omega_c t + \theta) dt.$$

- **Step 2:** Evaluate the integral.

$$\bar{x} = \lim_{T \rightarrow \infty} \frac{A}{2T\omega_c} [\sin(\omega_c T + \theta) - \sin(-\omega_c T + \theta)].$$

- **Step 3:** Limits.

As $T \rightarrow \infty$, the denominator $2T$ grows while the numerator is bounded between $-2A$ and $2A$.

- **Result:** $\bar{x} = 0$. Since this equals the ensemble mean $E[X(t)] = 0$, the process is **mean-ergodic**.

5) List of Important Formulas

Formula / Concept	Result
Mean Function	$\mu_X(t) = E[X(t)]$
Autocorrelation	$R_X(t_1, t_2) = E[X(t_1)X(t_2)]$
Autocovariance	$C_X(t_1, t_2) = R_X(t_1, t_2) - \mu_X(t_1)\mu_X(t_2)$
Correlation Coefficient	$\rho_X(t_1, t_2) = \frac{C_X(t_1, t_2)}{\sqrt{C_X(t_1, t_1)C_X(t_2, t_2)}}$
WSS Autocorrelation	$R_X(\tau) = E[X(t)X(t + \tau)]$
Time Average (Mean)	$\bar{x} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt$